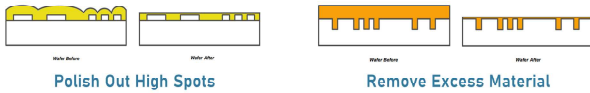


1 What is CMP

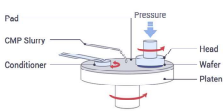
CMP = Chemical-Mechanical Planarization

Purpose of CMP

Using a combination of chemistry and abrasion under controlled conditions to remove specific amounts of material from the wafer surface and maintains precision at ever-decreasing wafer device nodes

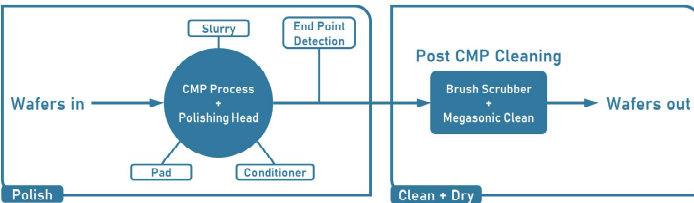


Components of CMP



Components	Purpose
Platen	Provide required rotation speed to the pad
Pad	To facilitate chemical reactions and accommodate abrasive particles
Polishing Head	Holding a wafer under pressure Rotating and sweeping it back and forth over the spinning platen with slurries or fluids
Slurry	Abrasive Particles provide mechanical polishing action Chemical Agents promote surface reactions
Conditioner	To optimize the pad profile

CMP Working Process



2 What is EPD

EPD = End Point Detection

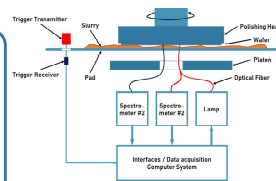
Purpose of EPD

Endpoint, the moment when the CMP process has removed the desired thickness of material. EPD, is to determine and control when the targeted amount of film has been removed across a wafer

Types of EPD

1 Optical

The optical endpoint detection method relies on the principle of optical reflectance. During the CMP process, a light source (typically a laser or LED) is directed onto the wafer surface. As the polishing progresses, the material being removed changes, causing variations in the reflective properties of the surface.



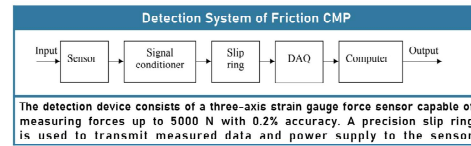
Components	Purpose
Light sources	A laser or LED is used as the light source, typically operating in the visible or near-infrared spectrum
Optical fibers	Guide the light from the source to the wafer surface and collect the reflected light
Spectrometer	Such as photodiodes or avalanche photodiodes, are used to convert the reflected light into electrical signals for further processing
Computer System	Signal Processing Unit: The electrical signals from the photodetectors are processed using analog-to-digital converters and digital signal processing algorithms to extract the endpoint information Control System: Integrates the optical endpoint detection system with the CMP tool, allowing for real-time monitoring and control of the polishing process

Optical EPD Working Progress	
Optical Module Design	To emit light onto the wafer surface during the polishing process The light is reflected off the polish surface and reference surface, which are adjacent to each other
Signal Generation	The reflected light from both surfaces is detected and corresponding signals are derived for the reflected light from the polish surface and the reference surface, respectively
Signal Processing	The signals are fed into an electronic system, where they are processed to determine the endpoint of the polishing process The endpoint is typically detected as a function of both signals, often using a difference function
Endpoint Detection	The endpoint is determined by analyzing the changes in the reflected light, which indicate the material removal rate The endpoint is typically detected when a predetermined difference signal is measured

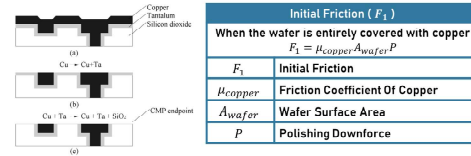
Different Methods of Optical EPD	
Threshold-based Detection	The endpoint is detected when the reflected light intensity crosses a predefined value
Derivative-based Detection	The derivative of the reflected light intensity is calculated, and the endpoint is detected when the derivative exceeds a certain threshold, indicating a significant change in the polishing rate
Pattern Recognition	Advanced pattern recognition algorithms, such as neural networks or support vector machines, are trained on historical data to identify the endpoint based on the reflected light intensity patterns
Multivariate Analysis	Multiple parameters, such as the reflected light intensity, polishing time, and other process variables, are analyzed together to improve the accuracy and robustness of endpoint detection

2 Friction

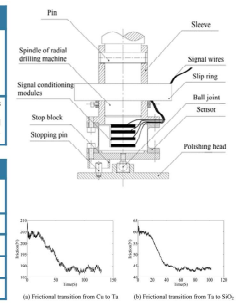
The friction-based EPD system detects changes in friction during the CMP process, which indicates the transition from one material layer to another, particularly for copper CMP. For example, the frictional force varies as the polished material changes from copper to the Ta barrier and then to silicon dioxide (SiO₂) dielectric.



The detection device consists of a three-axis strain gauge force sensor capable of measuring forces up to 5000 N with 0.2% accuracy. A precision slip ring is used to transmit measured data and power supply to the sensor



Initial Friction (F_1)	
When the water is entirely covered with copper	
$F_1 = \mu_{copper} A_{wafer} P$	
F_1	Initial Friction
μ_{copper}	Friction Coefficient Of Copper
A_{wafer}	Wafer Surface Area
P	Polishing Downforce

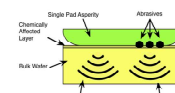


Friction at Copper-Ta Interface (F ₂)	
When the interface of copper and Ta barrier is polished	
$F_2 = \mu_{barrier} A_{barrier-1} P + \mu_{copper} A_{copper} P$	
The frictional change	
$\Delta F_1 = F_1 - F_2 = (\mu_{copper} - \mu_{barrier}) A_{barrier-1} P$	
F_2	Friction at Copper-Ta Interface
ΔF_1	The Frictional Change
$\mu_{barrier}$	Friction Coefficient of Ta, Tantalum
$A_{barrier-1}$	Initial Surface Area of Ta, Tantalum

Friction at Ta-SiO ₂ Interface (F ₃)	
When the interface of Ta barrier and SiO ₂ dielectric is polished	
$F_3 = \mu_{barrier} A_{barrier-2} P + \mu_{copper} A_{copper} P + \mu_{dielectric} A_{dielectric} P$	
The frictional change	
$\Delta F_2 = F_2 - F_3 = (\mu_{barrier} - \mu_{dielectric}) A_{dielectric} P$	
F_3	Friction at Ta-SiO ₂ Interface
ΔF_2	The Frictional Change
$\mu_{dielectric}$	Friction Coefficient of SiO ₂
$A_{barrier-2}$	Polished Surface Area of Ta, Tantalum
$A_{dielectric}$	Surface Area of SiO ₂

3 Acoustic Emission

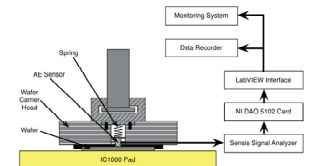
The acoustic emission endpoint detection relies on the propagation of elastic waves created as acoustic energy released in a solid during plastic deformation. There are 3 types of AE sources during CMP process, by filtering out the unnecessary ones, we can detect the CMP progress on time.



Acoustic Emission	
Primary source	The plastic deformation created by the interactions between slurry abrasives and the chemically weakened wafer surface
Secondary sources	Elastic contact between individual pad asperities and wafer surface
Other sources	Friction, Rubbing, Deformation

Acoustic Emission EPD Working Progress	
Sensis Signal Analyzer	AE signal is preamplified by 50 db, filtered the signal through a high pass filter with cutoff frequency of 50khz
NI system	Record the friction force data form the built in sensor
Labview	Filtered signal go through a RMS (root mean square) filter with 1ms time constant
	Remove high frequency noise and sudden changes in signal, which reduce false triggering by the first calculating an RMS average over a 3s window of data
	The data was then applied a moving average on the resulting RMS data

Moving Average RMS Equation	
$Y_i = \frac{1}{M+1} Y_i + \frac{1}{M} \sum_{j=i-M}^{i-1} Y_j$	
Y_i	The New Weighted Average RMS Value
Y_j	Current RMS Data Point
M	Number Of Previous RMS Data Points To Average

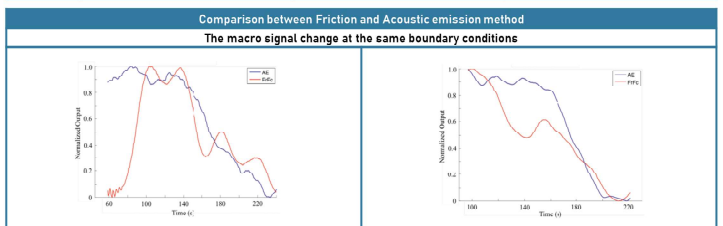


3 Conclusion

Comparison

we will discuss the advantage, limitation of each EPD methods above

Comparison between each EPD method above		
	Optical	Friction
Principle	Detect changes in the reflectance of the wafer surface during polishing by optical sensors	Monitors the change in friction between the polishing pad and the wafer during polishing
Advantages	High accuracy and resolution	Real-time monitoring
	Non-contact and non-destructive	Simple and cost-effective
		Capable of detecting subtle changes in friction corresponding to different material interfaces
Limitations	Affected by variations in slurry composition and wafer surface conditions	Less accurate than optical methods
	Requires precise calibration to setup	Difficult to integrate the AE into CMP process, due to the sensor must be contact to the source, to prevent signal from damping
	Expensive and complex to implement	Due to the rotational nature of CMP process, may affect the EPD accuracy



References		
Optical	Friction	Acoustic emission
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